

# Rebuttal for CoRL Submission #2468

We thank the Area Chair and reviewers. We respond below to R1–ZQwd, R2–jAyD, and R3–Fcwd. The new evidence directly addresses the requested progress baselines, optimization ambiguity, mask scope, component ablations, and evaluation protocol. We will revise the paper to narrow the claim to an online, tactile-supervised short-horizon contact prior that reduces pre-contact stall.

**[AC, R1, R3] Is VPCE only task progress, and how does it compare with phase/hierarchical conditioning?** We thank the reviewers and AC for highlighting this valuable comparison. We added phase-4 and normalized progress-20 predictors on the same held-out trajectories. Table I shows that VPCE consistently improves contact-prediction AUC/F1 over both alternatives. Thus trajectory time explains part of contact timing, but current wrist RGB and robot motion contain additional contact-specific information.

**TABLE I**  
CONTACT-PREDICTION AUC/F1.

| Task   | Phase     | Prog.     | VPCE             |
|--------|-----------|-----------|------------------|
| SWIPE  | .670/.000 | .853/.597 | <b>.963/.759</b> |
| WIPE   | .666/.000 | .743/.406 | <b>.885/.705</b> |
| INSERT | .684/.000 | .808/.539 | <b>.946/.735</b> |

line with a state-dependent 15-step horizon.

**[R1, R3] Does the teacher transfer, and how much target tactile data is required?** The source VPCE teacher is trained on the source dataset contains 639 (545/94) complete tactile trajectories. For downstream target-domain training, we use 60 tactile trajectories each for WIPE/SWIPE/INSERT, respectively. On trajectory-disjoint held-out data, the target-domain teachers obtain AUC/F1 .912/.731, .960/.774, and .940/.725. Thus the source teacher covers diverse contact regimes, while approximately 60 target trajectories suffice to generalize to unseen trajectories in each downstream domain.

**[R2] Are the learned loss weights well posed?** The original design emphasizes sparse but control-critical contact-transition samples relative to abundant free-space samples. However, because  $\gamma$  and  $\delta$  are optimized through the same weighted denoising loss, the model can reduce the overall objective by directly lowering the loss weights without improving denoising. We inspected checkpoints across tasks and found that this behavior is task dependent: at mean contact  $\bar{c}=1$ , the effective weights for both visual (V) and visuo-tactile (VT) policies are approximately 1.005, 2.1, and 4.4 on SWIPE, WIPE, and INSERT, respectively. This confirms that the learned formulation is not well posed. We therefore replace the learned term with the fixed schedule  $w(\bar{c}) = \text{clip}(1 + \alpha\bar{c}, 1, 1 + \alpha)$ , where  $\alpha$  is a preselected hyperparameter and receives no gradients.

**[R2] Does Eq. (9) mask proprioception, and how is free-space motion smooth?** Eq. (9) is intended as contact-conditioned *tactile routing*, not removal of robot state. We split the low-dimensional block as  $\tilde{f}_{\text{low}} = [f_{\text{prop}}, f_{\text{tac}}]$  and

route it as  $\tilde{f}_{\text{low}} = [f_{\text{prop}}, m(\bar{c})f_{\text{tac}}]$ : TCP state and gripper width are always retained, while only tactile/contact channels are suppressed when predicted contact is low. Thus free-space control still uses wrist RGB, proprioception, the VPCE horizon, action chunks, and receding-horizon execution. The submitted SWIPE ablation drops from 9/10 to 0/10 without contact masking, supporting the importance of the contact-conditioned masking pathway; the clarified tactile-only routing obtains 13/20. We will revise Eq. (9) to state this decomposition explicitly.

**[AC, R3] Why retain gating if success is unchanged?** Although gating does not increase the final binary success rate, that metric does not capture trajectory quality. As shown qualitatively in the supplementary video, the gated policy enters contact more directly and maintains alignment, whereas the no-gating policy requires more corrective inference/action chunks before establishing aligned contact. Thus gating improves contact-entry trajectory quality by adaptively rebalancing visual and low-dimensional feedback. We will add this qualitative comparison and the feature-weight traces to the main text.

**[R2] Could operator-held props assist the policy?** After rollout start, the holder did not track the end effector, reposition the object, or assist alignment; all methods used the same initial placement, success criterion, and 60-s timeout. Importantly, this shared protocol did not substantially improve the non-VPCE baselines on SWIPE: DP-V and DP-VT achieved only 1/20 and 2/20 success, whereas VPCE-DP-V and VPCE-DP achieved 17/20 and 18/20. Thus, the reachable held setup alone does not explain the performance gap; with the predicted contact horizon, the policy more reliably commits through the final approach into contact and manipulation, directly addressing the pre-contact stall targeted by this work. To reduce sampling uncertainty and operator-held variance under the same protocol, we further expanded VPCE-DP to 30 trials per task, obtaining 28/30 on SWIPE (93.3%), 22/30 on WIPE (73.3%), and 18/30 on INSERT (60.0%).

**[R2, R3] Why exclude tactile input from VPCE, and how are labels and contact directions handled?** VPCE intentionally excludes tactile input because its role is to anticipate contact before touch is informative and to permit tactile-free deployment; tactile is used as privileged training supervision. In the full policy, tactile feedback remains available after contact. For the current planar tasks, contact labels use the temporal change of a base-frame  $z$ -force proxy. A threshold of 0.095 is used for the source, SWIPE, and INSERT data; WIPE uses 0.0507 due to its different sensor scale. Each threshold is selected once from the training force-change distribution and then fixed. A  $0.6\times-1.4\times$  threshold sweep changes the 15-step positive-label rate smoothly (SWIPE .296  $\rightarrow$  .217, WIPE .582  $\rightarrow$  .379, INSERT .390  $\rightarrow$  .290), rather than creating labels only at one tuned value. Contacts with changing normals can

instead use  $\|\Delta \mathbf{F}\|_2$ , local normal/shear changes, hysteresis, or learned change-point labels without changing the VPCE-policy interface.

**[R1, R3] What is the main contribution relative to predictive force guidance and tactile representation learning?**

We will sharpen the claim: the contribution is the identification of pre-contact stall, tactile-supervised distillation of a compact future-contact horizon into visual-proprioceptive inputs, and evidence that this signal can initiate contact even without tactile sensing at deployment. The individual gates and loss weights are not claimed as the primary novelty. Unstructured tactile encoders such as Sparsh are complementary for representing post-contact feedback; comparing them does not replace the question addressed here—how to act before tactile observations become informative. We will expand related work and state the missing Sparsh/full hierarchical policy comparisons as evaluation limitations.